Welcome to
MAT 103

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Textbook

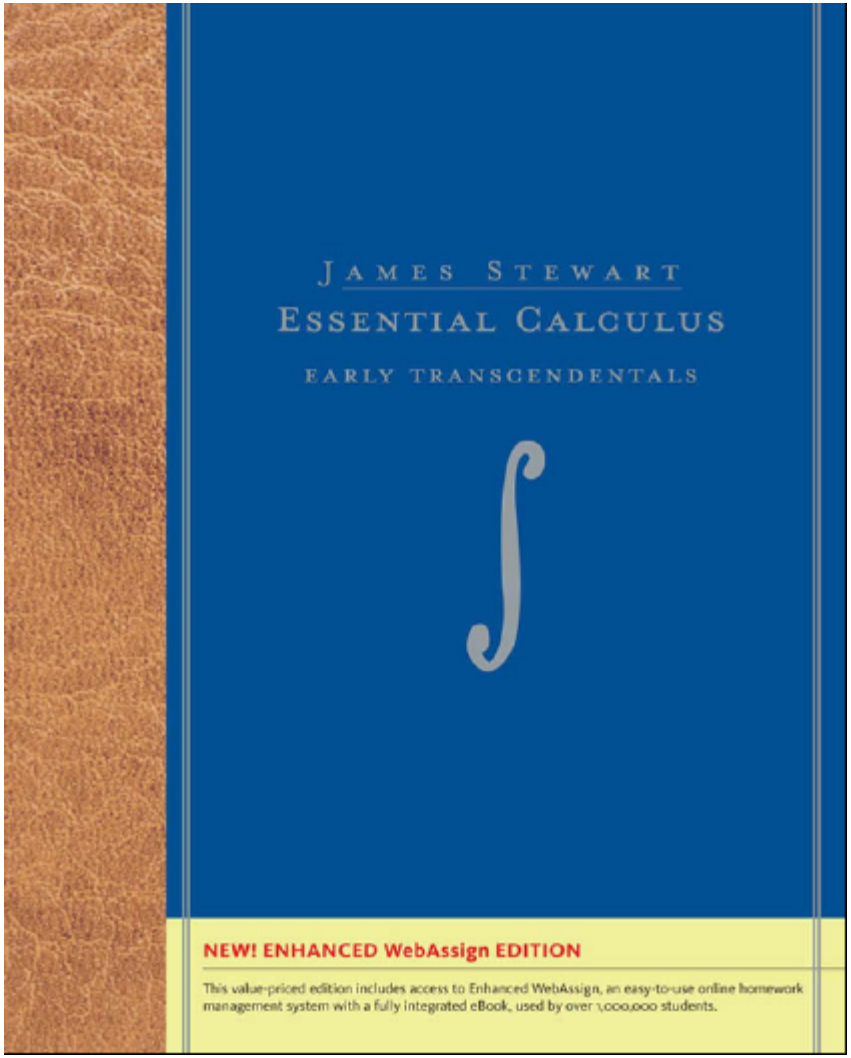
ESSENTIAL CALCULUS

EARLY TRANSCENDENTALS

JAMES STEWART

McMaster University and University of Toronto

Textbook looks like this



Easily avaialbe online

More about syllabus: **<u>Chapter 8 to 13</u>**

- 1 FUNCTIONS AND LIMITS
- 2 DERIVATIVES
- 3 INVERSE FUNCTIONS: Exponential, Logarithmic, and Inverse Trigonometric…
- 4 APPLICATIONS OF DIFFERENTIATION
- 5 INTEGRALS
- 6 TECHNIQUES OF INTEGRATION
- 7 APPLICATIONS OF INTEGRATION
- 8 SERIES
- 9 PARAMETRIC EQUATIONS AND POLAR COORDINATES
- 10 VECTORS AND THE GEOMETRY OF SPACE
- 11 PARTIAL DERIVATIVES
- 12 MULTIPLE INTEGRALS
- 13 VECTOR CALCULUS

Highlight of this course:

- You are going to do a lot of self study and teach yourself and teach each other.
- I am not going to teach (like on black-board or slides); I will just be here as a coordinator
- Aim is to promote and reward self-learning.

Instruction Plan

- <u>Pre-Lecture Activity</u>
  - Each lecture will be mapped to a pre-planned portion of the syllabus.
  - Students will be required to study these material and come to class prepared to
    - Answer basic conceptual questions from that material
    - Ask at least two basic question from that section to a fellow classmate.

- **Lecture Activity**
  - Students will form pairs
  - On a blank sheet of paper, each student will write question on which he wishes to evaluate his parter.
  - The beginning 20 minutes should be used in oral discussion, feedback, teaching each other.
  - Then 15 minutes should be used in writing detailed solution of asked problem.
  - Last 10 minutes to be spent on evaluation of written solution and assigning marks.
  - Last 5 minutes will go into recording marks and uploading picture of evaluated paper.

Assessment

- Assessment by other student 30 %
- Quality evaluation of other student 30 %
- Midterm (traditional format) 20 %
- Final exam (traditional format) 20 %

Lets get started

- **Form pairs:** Pair with the person on your left if he/she is alone. Else “hook up” with the person on your right.

- Take out a sheet of paper. Write your name and your partner’s name on it.

- Write two Calculus question on your sheet: Should be simple conceptual: like “whats a derivative”, “What is integration by parts”, etc

- (**20 minutes Oral discussion**): If your partner answers correclty, well and good. Else, teach him/her what you think is the correct answer.

- (10 minutes): Write detailed solution of the problem posed by your partner

- (5 minutes): Give marks to your partner out of 10.

- (Last five minutes) Click a picture of your sheet, upload them to prescribed place.